

VON ANDREW M. CASTILLO

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I am a detail-oriented Computer Science student at the University of Makati with hands-on software development experience across full-stack applications, game mechanics, and backend systems.

During my internship at TBL Hub, I architected full-stack web features and optimized C# RESTful API routes, successfully reducing API query latency by 20% for the UMak Store platform. I also spearheaded end-to-end Quality Assurance (QA) testing, bug tracking, and validation for the Bantay Bayan application, ensuring high feature reliability prior to production releases.

TECHNICAL SKILLS

- **Programming Languages:** C#, Java, JavaScript, Python, SQL, HTML5, CSS3
- **Backend & API Development:** RESTful APIs, Node.js, Express, Flask,
- **Databases & Cloud Storage:** SQL (MySQL/PostgreSQL), Firebase Firestore, Cloudinary
- **Frameworks & Engines:** Unity Engine, React, Next.js, Bootstrap, Tailwind CSS
- **Hardware & IoT:** NodeMCU, Arduino, Microcontroller Socket Interfacing
- **Tools & Platforms:** Git, GitHub, Visual Studio, VS Code, PyCharm, Postman

EXPERIENCE

Software Developer Intern | *TBL Hub* (February 2026 – May 2026)

- Developed full-stack web features and optimized API routes for the UMak Store platform.
- Engineered logic and form structures for the UMak Academic Questionnaire System.
- Conducted Quality Assurance (QA) testing, bug tracking, and functionality validation for the Bantay Bayan application.
- Completed 486 hours of technical internship focusing on agile collaboration and software delivery.

PROJECT SHOWCASE

Back-End Developer/Database Developer | *RainHarvest* — IPOPHL Innovation Entry (May 2026 - June 2026)

- Developed an AI-driven, climate-resilient solution for the Intellectual Property Office of the Philippines (IPOPHL) competition focused on inclusive community technology.
- Engineered backend API services and database structures to process real-time environmental and community data.
- Integrated machine learning classification logic to deliver actionable, low-latency insights for localized decision-making.
- Designed an accessible, low-bandwidth user interface tailored for reliable operation across underserved local communities.

Game Developer | *Scamester* (August 2025 - May 2026)

- Engineered REST API bridges connecting machine learning classification endpoints directly with live C# Unity game logic.
- Established Firebase Firestore datastore integration for dynamic session tracking and user synchronization.
- Implemented responsive UI state handlers and core game mechanics inside the Unity Engine.

Back End Developer | *Booklat* (February 2025 - June 2025)

- Architected C# backend logic and transactional database schemas for a digital library repository.
- Integrated Cloudinary API pipelines for dynamic cover image compression, upload management, and cloud storage.
- Configured real-time datastore listeners and security rules in Firebase.

Game Developer | *Deception* (August 2024 - January 2025)

- Programmed core gameplay mechanics, event trigger routines, and state handlers in C# and Unity.
- Integrated dynamic character animation states, environmental FX, sound management systems, and audio queues.
- Enforced strict game design sequences and execution parameters across all level triggers.

Full Stack Developer | *Mobile Pet Feeder* (February 2024 - June 2024)

- Fabricated an automated physical hardware prototype using NodeMCU microcontrollers and Arduino boards.
- Developed a native Java mobile interface communicating wirelessly with hardware microcontrollers via network sockets.
- Programmed backend trigger routines and error-handling mechanisms across microcontroller and mobile app layers.

Back End Developer | *Library Management System* (August 2023 - January 2024)

- Designed normalized relational database schemas and indexed table structures using SQL.
- Programmed desktop administrative application logic using C#.
- Tuned SQL query parameters and execution paths, significantly increasing data retrieval speed.

EDUCATION

Bachelor of Science in Computer Science | University of Makati, Philippines (2022 - 2026)

Senior High School Diploma in TVL – ICT | Benigno “Ninoy” Aquino Senior High School, Philippines (2020 - 2022)

CERTIFICATES

IT Specialist - Databases | Certiport (December 2025)